Perception-Constrained Altitude Control for Aerial–Ground Cooperative Tracking with a Shared Downward Camera

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***Abstract*— We present a genuinely-controlled, reproducible software-in-the-loop (SITL) study of aerial–ground cooperative tracking in which a single downward UAV camera is the ground vehicle's *sole* exteroceptive sensor, serving simultaneously as its navigation reference (path segmentation) and its state estimator (ArUco/EKF). We show this dual role induces a fundamental altitude tension: low altitude sharpens estimation (ground-sampling noise $R \propto z^2$) but starves navigation coverage (the path lookahead leaves the camera footprint), while high altitude does the reverse. Our contributions are: (i) we formalize the coverage–resolution tension and derive the perception-constrained altitude policy — fly as low as segmentation allows, climbing anticipatorily into turns; (ii) we characterize the resulting cost tradeoff in reproducible SITL; and (iii) we show a perception-driven event trigger is the computational enabler of the heavier belief-space planner, cutting online planner solves $\approx 28\times$ ($1255\pm9 \rightarrow 45\pm1$ per run) and total planner CPU $\approx 14\times$ at matched estimation quality (covariance 0.0063, 100% visibility for both continuous and event-triggered planning), trading a small amount of trajectory smoothness for the compute saving. We are explicit about scope: the isolated observability benefit is *not* separable on this platform because the shared camera couples navigation and estimation, and the low-altitude/high-speed corner is non-reproducibly unstable — both stated as findings. We also report a methodological point: headless execution is a prerequisite for reproducible closed-loop active-perception study (identical-configuration trajectories fork by ≈ 2.7 m with the Gazebo GUI versus ≈ 0.01 m headless). Throughout we mark which properties are proven offline, implemented, shown in simulation, or future work.**

I. INTRODUCTION

The deployment of Unmanned Ground Vehicles (UGVs) in unstructured and GPS-denied environments is heavily constrained by their limited perception horizons. Ground-level sensors such as LiDAR and stereo cameras suffer from occlusion, narrow fields of view (FoV), and an inability to anticipate upcoming path curvature. To mitigate this, aerial–ground cooperative systems employ an Unmanned Aerial Vehicle (UAV) as an elevated "eye in the sky" for a perception-limited UGV.

Most cooperative-tracking frameworks treat the UAV as a *passive* observer: it follows a predefined trajectory or simply minimizes distance to the UGV, treating the UGV's state estimation as a downstream byproduct [3]. This decoupling is fragile: when the UGV maneuvers aggressively or is occluded, the UAV can lose the target, the estimation covariance grows, and tracking degrades. Active-perception planning, where the sensor trajectory is optimized to maximize information gain [4], addresses part of this but typically relies on continuous, computationally expensive replanning and rarely integrates robust tracking guarantees for the aerial vehicle.

This paper studies a *bidirectional* observability-aware architecture in which *both* vehicles adapt to the ground vehicle's estimation uncertainty. Its contributions, as realized in the accompanying open implementation, are:

- **Formalization of the single-camera coverage–resolution tension** and the perception-constrained altitude policy: with one downward camera serving both the ground vehicle's navigation (path segmentation) and its state estimate (ArUco/EKF) [5], [6], low altitude sharpens estimation ($R \propto z^2$) but starves navigation coverage; the optimum flies as low as segmentation allows, climbing anticipatorily into turns. [PROVEN-OFFLINE]
- **A genuinely-controlled, reproducible SITL realization** in ROS 2/Gazebo (vision/ArUco pipeline, view-quality-coupled EKF, belief-space UGV MPC, and the UAV planner/ancillary pair [1], [7]), including a methodological finding: headless execution is a prerequisite for reproducible closed-loop active-perception study (≈ 2.7 m vs. ≈ 0.01 m identical-configuration trajectory fork). [IMPLEMENTED]
- **A validated co-design result**: the perception-driven event trigger is the computational enabler of the heavier belief-space planner, cutting online planner solves $\approx 28\times$ and planner CPU $\approx 14\times$ at matched estimation quality, for a small loss of inter-trigger trajectory smoothness. [SIMULATION]

We are explicit about scope: the *isolated* observability benefit is not separable on this platform because the shared camera couples navigation and estimation, and the binding low-altitude/high-speed corner is non-reproducibly unstable — both stated as findings (Sec. VII), not hidden.

We deliberately separate the idealized robust-MPC theory (offline mRPI, tightened constraints, terminal-set recursive feasibility) from what the code enforces online, and we validate the computational claim in simulation. The remainder of the paper is organized as follows. Section II reviews related work; Section III states the problem; Section IV gives the vehicle and perception models; Section V develops the coupled event-triggered tube-MPC; Section VI states the theoretical properties and the exact implementation status; and Section VII reports the simulation study.

II. RELATED WORK

Robust and tube MPC. Tube MPC [1] decomposes control into a nominal plan and an ancillary law that confines

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the true state to a robust positively invariant (RPI) tube; Raković et al. [2] give the standard outer approximation of the minimal RPI set by a truncated Minkowski series. These guarantees assume a fixed, bounded disturbance set and constraint tightening computed offline. Time-varying (homothetic/parameterized) tubes generalize the fixed tube but require the scaling to enter the optimization with its own feasibility argument; our online adaptivity (Section V) is a lighter-weight surrogate and we mark the homothetic treatment as future work.

Perception-aware and active planning. PAMPC [4] optimizes quadrotor trajectories for perception objectives. Belief-space planning [5], [6] augments the planning state with the estimator covariance. Our belief-space UGV controller is in this family. The *bidirectional* coupling — the *target* shaping its trajectory to stay observable while the *tracker* adapts its robustness to perception quality — is the design novelty of the architecture. What we *evidence* in this paper is narrower and concrete: the integrated SITL realization of that architecture and the perception-driven trigger's computational result (Sec. VII). Isolating the coupling's observability benefit is future work requiring the ground vehicle's navigation to be decoupled from the tracking camera (Sec. VI).

Event-triggered MPC. Event-triggered schemes [7] resolve only when a state-dependent condition fires. We instantiate a perception-driven trigger with a critical tube-exit condition plus anticipatory conditions.

III. PROBLEM FORMULATION

A UAV tracks a UGV in a global frame $\mathcal{F}_W$. The UAV carries the only exteroceptive sensor: a downward monocular camera that detects an ArUco marker on the UGV, yielding a noisy pose measurement whose quality depends on the imaging geometry. An EKF fuses these detections with UGV wheel odometry to produce a belief (μ, Σ) over the UGV state. The objective is to (i) keep the UGV inside the camera footprint so that Σ stays bounded, (ii) track the UGV robustly under bounded disturbances, and (iii) do so while limiting the number of online optimizations. The UGV must track a ground lane while remaining observable.

IV. SYSTEM ARCHITECTURE AND MODELS

Fig. 1 shows the implemented system and its signal flow.

A. UGV dynamics

The UGV uses unicycle kinematics with first-order actuator lag. Let $\mathbf{x}_g = [x_g, y_g, \theta, v, \omega]^T$ be the state, (x_g, y_g) the position, θ the heading, (v, ω) the linear/angular velocities, and $\mathbf{u}_g = [v_c, \omega_c]^T$ the commanded velocities. The discrete-time dynamics are

$$\mathbf{x}_{g,k+1} = \mathbf{f}_g(\mathbf{x}_{g,k}, \mathbf{u}_{g,k}) + \mathbf{w}_g, \tag{1}$$

where the actuator lag is governed by time constants τ_v, τ_ω and $\mathbf{w}_g$ is process noise. [IMPLEMENTED]

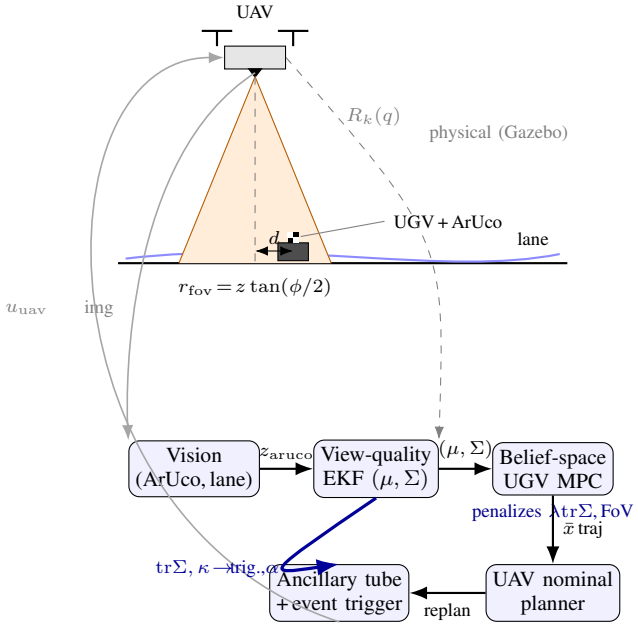


Fig. 1. Implemented system and signal flow. A downward-camera UAV (footprint radius $r_{\text{fov}} = z\tan(\phi/2)$) observes an ArUco-tagged UGV at horizontal offset d. **Black:** forward data flow (vision→EKF→belief-space UGV MPC→UAV nominal planner→ancillary tube/event trigger→UAV); **blue:** the perception→control couplings actually used — the UGV cost penalizes $\lambda\,\text{tr}\Sigma$ and FoV exit, and the EKF covariance/curvature drive the event trigger and tube radius α; **dashed grey:** the modeled dependence of measurement noise $R_k(q)$ on UAV geometry. All arrows are implemented ROS 2 signals; the figure asserts no performance claim.

B. UAV model

Although the platform is a quadrotor, the planner and the ancillary tracker treat it as a **velocity-controlled point mass** (single integrator); an inner-loop attitude/thrust controller, provided by the Gazebo velocity-control plugin, is assumed to realize commanded velocities. With position $\mathbf{p} = [p_x, p_y, p_z]^T$ and velocity command $\mathbf{u} = [v_x, v_y, v_z]^T$,

$$\mathbf{p}_{k+1} = \mathbf{p}_k + \Delta t\,\mathbf{u}_k + \mathbf{w}_k, \qquad A_d = I_3,\ B_d = \Delta t\, I_3, \tag{2}$$

with bounded disturbance $\mathbf{w}_k \in \mathbb{W}$. All three axes are fully actuated: v_z is a genuine decision variable and the altitude policy of Sec. IV-D is realized closed-loop in SITL (the UAV altitude spans $[3, 8]$ m; Sec. VII). A second-order double-integrator (acceleration input) is the more faithful rigid-body abstraction; we use (2) because it is exactly what the implementation optimizes. [IMPLEMENTED]

C. View-quality-dependent perception

The camera's vantage point affects estimation accuracy. We use a differentiable view-quality metric $q_k \in (0, 1]$ that degrades as the UGV nears the FoV edge and as the altitude departs from a precision-optimal value z_{opt} (too low loses footprint; too high loses pixel resolution). With horizontal UAV–UGV distance $d = \|\mathbf{p}^{xy} - \mathbf{p}_g\|$ and footprint radius

$r_{\mathrm{fov}}(z) = z\tan(\phi/2)$,

$$q_k = \underbrace{\frac{1}{1+e^{\beta(d-r_{\mathrm{fov}})}}}_{\text{FoV gate}} \cdot \underbrace{e^{-d^2/(2\sigma_{xy}^2)}}_{\text{centering}} \cdot \underbrace{e^{-(z-z_{\mathrm{opt}})^2/(2\sigma_z^2)}}_{\text{altitude precision}}, \quad (3)$$

with $\sigma_{xy} = r_{\mathrm{fov}}/2$. The altitude factor makes q_k *non-monotone* in z (it does not tend to 1 as $z \to \infty$), removing the incentive to climb without bound. [IMPLEMENTED]

The measurement covariance grows as view quality drops. The implementation scales the baseline ArUco covariance R_{full} inversely with q_k,

$$R_k(q_k) = \frac{R_{\mathrm{full}}}{q_k + \epsilon}, \qquad \epsilon = 10^{-3}, \quad (4)$$

so $R_k \to R_{\mathrm{full}}$ for an excellent view and $R_k \to \infty$ (the update ignores the marker) as $q_k \to 0$. This couples UAV geometry directly to EKF performance. [IMPLEMENTED]

For the quantitative study we use the physically-grounded *ground-sampling-distance* (GSD) noise model: pixel-level localization is fixed while the GSD (m/px) grows linearly with altitude, so the metric position-measurement covariance grows *quadratically*,

$$R(z) = R_{\mathrm{full}}\,(z/z_0)^2\left(1 + (d/r_{\mathrm{fov}})^2\right), \quad (5)$$

with a mild FoV-edge inflation $(1 + (d/r_{\mathrm{fov}})^2)$. Unlike the Gaussian-bowl form (4) — which is non-monotone in z and instead models resolution dropout away from a nominal altitude — (5) is monotone increasing in z, matching the theory's resolution assumption and the $R \propto z^2$ side of the coverage–resolution tension (Sec. IV-D). Equation (5) is the measurement-noise model used in all reported results. [IMPLEMENTED]

D. Segmentation coverage model

The same downward camera also supplies the UGV's *navigation* reference: an OpenCV segmenter extracts the lane and seeds a waypoint a lookahead distance ahead of the vehicle. That waypoint must lie inside the camera footprint or segmentation fails and the UGV loses its reference. With footprint radius $r_{\mathrm{fov}}(z) = z\tan(\phi/2)$ and required lookahead $d_{\mathrm{look}}(\kappa)$ — which grows with local path curvature κ, since a curving lane exits the footprint sooner and needs more forward view — define the non-dimensional *coverage factor*

$$\eta \;=\; \frac{r_{\mathrm{fov}}(z)}{d_{\mathrm{look}}(\kappa)} \;=\; \frac{z\tan(\phi/2)}{d_{\mathrm{look}}(\kappa)}, \quad (6)$$

with segmentation feasible for $\eta \gtrsim 1$. This is the crux coupling: the estimator wants *low* altitude ($R \propto z^2$, better resolution), while (6) demands *sufficient* altitude for navigation. The estimation-optimal altitude is thus the lowest that keeps $\eta \gtrsim 1$, and this floor rises with curvature — the analytic basis for climbing into turns. The aerial planner (Sec. V-B) enforces $\eta \gtrsim 1$ as a soft lower bound; Sec. VII reports the empirical cost tradeoff over η. [IMPLEMENTED]

V. Coupled Event-Triggered Tube MPC

The estimation uncertainty Σ_k enters the control formulations of both vehicles.

A. Information-aware UGV control (belief-space NMPC)

The UGV controller augments the planning state with the covariance, $\xi_k = (\mu_k, \Sigma_k)$, and *propagates both through the horizon*. At each predicted step it forms the no-observation prediction $\Sigma_{\mathrm{pred}} = F\Sigma F^T + Q$ and a post-measurement covariance Σ_{obs} (Joseph update with R_k from (4)), and blends them by a smooth visibility gate $\gamma = 1/(1 + e^{\beta(d-r_{\mathrm{fov}})})$:

$$\Sigma_k = (1-\gamma)\,\Sigma_{\mathrm{pred}} + \gamma\,\Sigma_{\mathrm{obs}}. \quad (7)$$

The cost minimizes tracking error while penalizing predicted uncertainty and FoV exit,

$$J_g = \sum_{k=0}^{N_g-1} \Big(\|\mathbf{e}_{g,k}\|_Q^2 + \|\mathbf{u}_{g,k}\|_R^2 + \lambda\ \mathrm{tr}(\Sigma_k) + w_{\mathrm{fov}}\, c_{\mathrm{fov}}(\mathbf{x}_{g,k}) \Big), \quad (8)$$

where the FoV term is a one-sided quadratic hinge $c_{\mathrm{fov}} = \big(\max(0,\, d_k - r_{\mathrm{fov}})\big)^2$ (a soft constraint, not an asymptotic barrier). The covariance rollout (7) makes (8) a nonconvex NLP, solved with SLSQP over $N_g = 5$. The two terms $\lambda\,\mathrm{tr}(\Sigma_k)$ and $w_{\mathrm{fov}} c_{\mathrm{fov}}$ are what make the controller belief-coupled and are absent from the passive baseline. [IMPLEMENTED]

B. Aerial nominal planner and altitude policy

The UAV computes a nominal trajectory $\bar{\mathbf{p}}$ over $N_a = 10$ by optimizing goal tracking, an altitude regularizer toward z_{opt}, and a visibility term that tolerates lateral offset only when altitude is sufficient:

$$J_a = \sum_k w_{\ell,k} \Big[\|\bar{\mathbf{p}}_k - \mathbf{g}\|^2 + w_z (z_k - z_{\mathrm{opt}})^2 + w_{\mathrm{alt}} \frac{\|\bar{\mathbf{p}}_k^{xy} - \mathbf{g}^{xy}\|^2}{z_k^2 + 0.1} \Big] + w_u \|\bar{\mathbf{u}}_k\|^2, \quad (9)$$

where $\mathbf{g}$ is the UGV's predicted end-of-horizon position (read from the belief-MPC rollout, one direction of the coupling) and $w_{\ell,k}$ up-weights later steps. The perception-constrained altitude policy of Sec. IV-D sets the altitude setpoint explicitly to the coverage floor plus an anticipatory margin,

$$z_k \;=\; \hat{d}_k / \tan(\phi/2) \;+\; \beta_\kappa\, \hat{\kappa}_k \;+\; z_{\mathrm{m}}, \quad (10)$$

where $\hat{d}_k$ is the estimated horizontal UAV–UGV offset (the first term keeps the target within the footprint), $\hat{\kappa}_k$ is the anticipated path curvature, and $\beta_\kappa \hat{\kappa}_k$ raises altitude into turns where the required lookahead $d_{\mathrm{look}}(\kappa)$ — and hence the coverage floor $\eta \gtrsim 1$ (6) — grows; z_{m} is a fixed margin that keeps the forward lookahead in view on straights. "Climb into turns to widen the footprint" is thus a derived consequence of the coverage constraint, not a hand-coded rule. [IMPLEMENTED]

C. Offline robust invariant set (tube) derivation

The ancillary law is $\mathbf{u}_k = \bar{\mathbf{u}}_k - K\mathbf{e}_k$ with tracking error $\mathbf{e}_k = \mathbf{p}_k - \bar{\mathbf{p}}_k$; substituting into (2),

$$\mathbf{e}_{k+1} = \underbrace{(A_d - B_d K)}_{=:A_K}\mathbf{e}_k + \mathbf{w}_k, \qquad \mathbf{w}_k \in \mathbb{W}. \quad (11)$$

Unlike standard tube MPC, we account for measurement noise $\mathbf{v}_k$ entering through the feedback gain; the effective disturbance is $\mathbf{w}_{\text{eff}} = \mathbf{w}_a - B_d K \mathbf{v}_k$, and for the offline computation $\mathbf{v}_k$ is bounded to a box (a hard, undocumented-confidence truncation of the otherwise Gaussian noise; see Section VI). For A_K Schur and $\mathbb{W}$ compact, the minimal RPI set is $\Omega_\infty = \bigoplus_{i=0}^{\infty} A_K^i \mathbb{W}$, which we outer-approximate by truncating after $N = 100$ terms and scaling the support values by $1/(1-\rho^{N+1})$, ρ the spectral radius of A_K, to conservatively bound the neglected tail [2]:

$$\Omega_N = \bigoplus_{i=0}^{100} A_K^i \mathbb{W}, \qquad \Omega \supseteq \Omega_\infty. \tag{12}$$

The state/input constraints are then tightened by the Pontryagin difference, $\bar{\mathbb{X}} = \mathbb{X} \ominus \Omega$, $\bar{\mathbb{U}} = \mathbb{U} \ominus K\Omega$. (12) and the tightening are computed offline and constitute a *certified robust bound for the idealized design*. [PROVEN-OFFLINE]

D. Online adaptive ellipsoidal tube

Propagating the polytope Ω online — vertex/support-function updates and a Pontryagin difference at every 20 Hz ancillary step — is impractical on the UAV's embedded companion processor. Rather than pay that overhead, the ancillary controller uses an *ellipsoidal* invariant-set surrogate

$$\mathcal{E}(\alpha_k) = \{\mathbf{e} : \mathbf{e}^T P_0 \, \mathbf{e} \le \alpha_k^2\}, \tag{13}$$

with a shape matrix P_0 (loaded from the offline design, else a Schur-consistent fallback) and a radius that adapts to the EKF horizontal covariance trace $\operatorname{tr}\Sigma_{xy} = \Sigma_{xx} + \Sigma_{yy}$:

$$\alpha_k = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) \operatorname{clip}\Big(\tfrac{\operatorname{tr}\Sigma_{xy} - \operatorname{tr}_{\min}}{\operatorname{tr}_{\max} - \operatorname{tr}_{\min}}, 0, 1\Big), \tag{14}$$

($\alpha_{\min} = 0.5$, $\alpha_{\max} = 2$). The tube *breathes*: it tightens when the estimate is sharp and loosens when perception degrades. We stress that (14) scales the *invariant-set membership/trigger radius* used online; it is *not* accompanied by an online re-derivation of (12) nor by re-tightening $\bar{\mathbb{X}}, \bar{\mathbb{U}}$. A homothetic tube with a matching recursive-feasibility argument is future work (Section VI). [IMPLEMENTED]Accordingly, α_k is a heuristic monitor/trigger radius, not a certified invariant radius. The certified objects are the *offline* mRPI set Ω (12) and the A_K-Schur ISS bound of Sec. VI (which rests on the contraction A_K and bounded $\mathbb{W}$, not on α_k). The choice $\alpha_{\min} < 1$ tunes *trigger sensitivity* (how eagerly a replan fires), not invariance; letting the membership radius drop below the nominal invariant size therefore does not weaken those certified guarantees or contradict the offline mRPI bound.

E. Perception-driven event trigger

The ancillary node runs at 20 Hz but requests a nominal replan only when one of four prioritized conditions fires (implemented literally as follows).

- **(1) Tube exit (critical, no lockout):** fire iff $\mathbf{e}_k^T P_0 \mathbf{e}_k > \alpha_k^2$, i.e. the state left the adaptive ellipsoid (13).
- **(2) Covariance spike:** $\operatorname{tr}\Sigma_{xy}(t) > \tau_{\text{cov}}$ and a sudden ($>1.5\times$) rise over ~ 0.5 s (e.g. ArUco occlusion).
- **(3) Curvature anticipation:** the discrete curvature of the UGV predicted path, $\hat{\kappa} = \max_i |\Delta\theta_i|/\Delta s_i$ — the wrapped heading change $\Delta\theta_i$ between consecutive predicted waypoints over their spacing Δs_i, maximized over the next few steps — exceeds $\tau_\kappa = 0.4$ rad/m, so the UAV climbs before the turn (raising the coverage floor of Sec. IV-D).
- **(4) Temporal drift:** elevated $\operatorname{tr}\Sigma_{xy}$ with time since last replan exceeding $T_{\min}$.

Conditions (2)–(4) carry a 1 s time-lockout (the chattering guard); condition (1) has none. Between triggers the ancillary law regulates the error via (11), and the previous nominal sequence is shifted forward.

We note that a purely fractional trigger on the *polytopic* tube, $\max(H_\Omega \mathbf{e}_k - \sigma h_\Omega) > 0$ with contracted set Ω_σ, is the natural formulation for the offline mRPI Ω and would furnish the tunable computation–accuracy parameter σ; the implementation instead uses the ellipsoidal test above, so we present the σ-polytopic trigger as future work rather than as an implemented result. [FUTURE WORK]

F. Nominal optimization actually solved

At each trigger the planner solves (9) over the single integrator (2) with velocity bounds $v_x, v_y \in [-1.5, 1.5]$, $v_z \in [-1, 1]$ m/s, warm-started from the previous solution (SLSQP, maxiter $= 15$). Altitude limits $z \in [z_{\min}, z_{\max}]$ are shaped by the cost; a hard terminal set $\mathbb{X}_f$ and the DARE terminal cost of the idealized formulation below are *not* enforced online. [IMPLEMENTED]

VI. Theoretical Properties and Implementation Status

We separate what is proven, what is inherited under assumptions, and what is only empirical.

Idealized recursive feasibility. For the error system (11) with Ω a true RPI set and tightened constraints $\bar{\mathbb{X}}, \bar{\mathbb{U}}$, a nominal MPC with a control-invariant terminal set $\mathbb{X}_f$ and DARE terminal cost is recursively feasible and the true state satisfies $\mathbf{x}_k \in \bar{\mathbb{X}} \oplus \Omega \subseteq \mathbb{X}$ [1], [2]. This holds for the *idealized* design; the code uses the ellipsoidal surrogate (13) and does not enforce tightening or a terminal set, so online recursive feasibility is *assumed, not verified*. [PROVEN-OFFLINE]

Bounded inter-trigger error (ISS). With A_K Schur and $\mathbf{w} \in \mathbb{W}$ bounded, (11) is input-to-state stable. For the implemented contraction $A_K = 0.6\, I_3$ this is immediate: $\|\mathbf{e}_{k+1}\| \le 0.6\|\mathbf{e}_k\| + \sup\|\mathbf{w}\|$, hence $\limsup_k \|\mathbf{e}_k\| \le \sup\|\mathbf{w}\|/0.4$. Thus, between triggers, the error stays bounded and the ellipsoid membership is a meaningful monitor. [PROVEN-OFFLINE]

Belief-MPC stability. Adding $\lambda \operatorname{tr}\Sigma_k$ and c_{fov} to the stage cost (8) makes it nonconvex and state-dependent; the standard terminal-cost machinery does not directly apply and we do not claim a formal stability proof for the UGV controller. [FUTURE WORK]

Stochastic/bounded mismatch. The mRPI machinery needs a bounded $\mathbb{W}$, whereas the EKF noise is Gaussian. The offline computation truncates $\mathbf{v}_k$ to a box without a

stated confidence level, and the online Σ is never mapped to a hard bound. A probabilistic/stochastic-tube treatment is future work. [FUTURE WORK]

Table I summarizes the claim status.

TABLE I

CLAIM STATUS: PROVEN OFFLINE VS. IMPLEMENTED VS. FUTURE

Claim	Status
Belief NMPC propagates Σ, penalizes $\mathrm{tr}\,\Sigma$+FoV	implemented
View-quality-coupled R_k (4)	implemented
Non-monotone view quality in altitude (3)	implemented
Adaptive ellipsoidal tube radius (14)	implemented
Four-condition perception event trigger	implemented
Offline mRPI outer bound (12), tightening	proven (offline)
Bounded inter-trigger error / ISS	proven (offline)
σ-polytopic fractional trigger, Ω_σ	future work
Online mRPI / homothetic tube / re-tightening	future work
Terminal-set + DARE recursive feasibility online	future work
UGV belief-MPC formal stability proof	future work

Limitations and honest scope. The study is in simulation only; the UAV is a point-mass single integrator; the tube adaptivity is in the trigger/monitor radius, not a re-derived invariant set; the mRPI is offline and not wired into the online loop; and no online recursive-feasibility certificate is enforced. Most importantly, *the isolated observability benefit of perception-awareness is not separable on this platform*, for two reasons detailed in Sec. VII: (a) because the shared downward camera keeps the target near the optical center (mean UAV–UGV offset 0.06–0.12 m), the estimation covariance is dominated by the altitude-dependent measurement noise and carries no separable tracking-difficulty signal; and (b) the low-altitude/high-speed corner that would stress the coverage margin is non-reproducibly unstable. We stress that this residual non-reproducibility is confined to that binding corner: in the operating regime the closed loop *is* reproducible once run headless (identical-configuration trajectories agree to ≈ 0.01 m; with the Gazebo GUI, rendering contention with the ROS 2 executor forks them by ≈ 2.7 m). Establishing the isolated observability benefit is future work requiring the ground vehicle's navigation to be decoupled from the tracking camera (onboard odometry or a second exteroceptive sensor).

VII. SIMULATION STUDY

Setup. The system runs in Gazebo with ROS 2: a TurtleBot3-class UGV with an ArUco marker, an X500-class UAV with a downward camera, an OpenCV lane/marker pipeline, the 5-state EKF, the belief-space UGV MPC, and the UAV planner/ancillary pair. Per-step telemetry (pose, ground truth, $\mathrm{tr}\,\Sigma$, FoV distance, ArUco visibility, view quality, replan events, solve times) is logged by a common logger identical across configurations.

Reproducibility (a prerequisite, and a methodological finding). Closed-loop active-perception study requires reproducible runs. We found the ROS 2/Gazebo pipeline is *not* reproducible when run with the GUI: identical-configuration ground-truth trajectories fork by ≈ 2.7 m and ArUco visibility swings between 0 and 100%, because the GUI rendering thread contends with the ROS 2 executor and perturbs the closed perception–control–physics loop. Running *headless* (server-only) removes this contention: identical-configuration trajectories then agree to ≈ 0.01 m and ArUco visibility is consistently 100%. All results below are headless. We report this because it is an often-unstated prerequisite for reproducible closed-loop SITL evaluation.

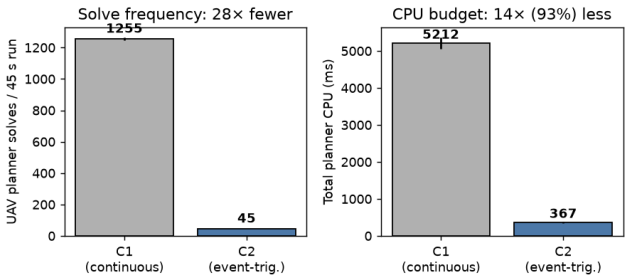


Fig. 2. Co-design (headless, R=3, nominal track): the perception-driven event trigger (C2) invokes the belief-space planner $\approx 28\times$ less often than continuous replanning (C1), cutting total planner CPU $\approx 14\times$ (93%) at matched estimation quality ($\mathrm{tr}\,\Sigma = 0.0063$, 100% visibility for both).

Co-design: event-triggering makes the belief-space planner affordable. The belief-space planner propagates the EKF covariance through its horizon and is comparatively expensive per solve. Continuous replanning (C1) spends a large, sustained CPU budget; the perception-driven event trigger (C2) invokes the *same* planner only when a covariance/curvature/drift condition fires. On the nominal track (headless, R=3), C2 reduces planner invocations $\approx 28\times$ ($1255 \pm 9 \rightarrow 45 \pm 1$) and total planner CPU $\approx 14\times$ ($5212 \pm 151 \rightarrow 367 \pm 20$ ms per 45 s run; Fig. 2), at matched estimation quality (steady-state $\mathrm{tr}\,\Sigma = 0.0063$ and 100% ArUco visibility for both). Per-solve time is *higher* for C2 (8.2 ± 0.2 vs. 4.2 ± 0.1 ms), consistent with a warm-start disparity — C1 re-solves every 50 ms from a near-identical previous solution, whereas each C2 solve starts from a larger state mismatch — but the $\approx 28\times$ reduction in solve count dominates, giving the net CPU saving. This is the co-design point: event-triggering makes the heavier perception-aware planner affordable on a constrained companion computer.

What event-triggering costs. It is not free: applying the previous nominal trajectory between triggers introduces small control discontinuities at trigger instants. We measure $\approx 20\times$ higher high-frequency trajectory content for C2 than C1 (RMS of the second position difference). The mean UAV–UGV offset is nonetheless smaller for C2 (0.055 ± 0.001 vs. 0.124 ± 0.003 m); we do *not* attribute this to event-triggering (the arms differ in planner cadence) and report it only for completeness. The honest summary is a standard event-triggered tradeoff: a large compute saving for a small loss of inter-trigger trajectory smoothness, at matched estimation quality.

The coverage–resolution altitude tension. With the GSD model enabled, we sweep altitude across three tracks of

TABLE II

CO-DESIGN METRICS, FIXED SYSTEM (HEADLESS, NOMINAL TRACK, MEAN $\pm$ STD OVER R=3). C1 = CONTINUOUS REPLANNING; C2 = PERCEPTION-DRIVEN EVENT-TRIGGERED. STEADY-STATE WINDOW $t \geq 8$ S. ESTIMATION QUALITY IS MATCHED; THE COMPUTE BUDGET IS NOT.

Metric	**C1 (Continuous)**	**C2 (Event-Trig.)**
Planner solves (# / run)	1255 ± 9	45 ± 1
Mean solve time (ms)	4.2 ± 0.1	8.2 ± 0.2
Total planner CPU (ms / run)	5212 ± 151	367 ± 20
Steady-state $\mathrm{tr}\,\Sigma$	0.0063	0.0063
ArUco visibility (%)	100	100
UAV–UGV offset (m)	0.124 ± 0.003	0.055 ± 0.001

increasing curvature (headless; low-altitude cells are the median of R=3 reps). Two costs move oppositely with altitude (Fig. 3): estimation uncertainty $\mathrm{tr}\,\Sigma$ *decreases* as altitude decreases (0.0094 at z=6 m to 0.0053 at z=2 m — the $R \propto z^2$ resolution gain, and track-independent), while the UGV steering effort needed to keep the path segmented *increases* as altitude decreases, and increases with path curvature. Hence the perception-constrained policy: fly as low as segmentation allows, climbing anticipatorily into turns where the navigation floor rises. In SITL the active-altitude policy realizes exactly this — the UAV altitude spans $[3, 8]$ m, climbing into turns (mean $\approx$5.8 m over high-curvature segments vs. $\approx$4.6 m on straights). [SIMULATION]

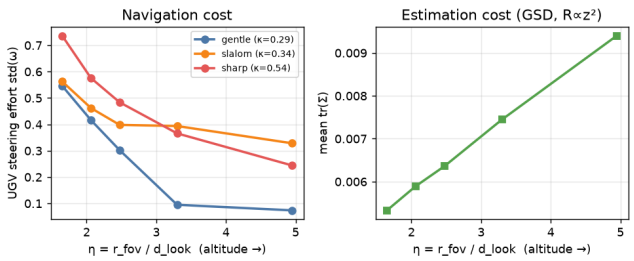


Fig. 3. The two jaws of the altitude tension ($\eta = r_{\mathrm{fov}}/d_{\mathrm{look}}$, increasing with altitude). *Left:* UGV steering effort (navigation cost) rises at low altitude and with path curvature. *Right:* estimation uncertainty $\mathrm{tr}\,\Sigma$ falls at low altitude (GSD, $R \propto z^2$; track-independent). The optimum flies as low as segmentation allows.

Scope, stated as findings. Two limits are findings, not hidden. *(a) The isolated observability benefit is not separable on this platform.* Because the single downward camera serves both navigation and estimation, and the tracker holds the target near the optical centre, $\mathrm{tr}\,\Sigma$ is dominated by the altitude-dependent measurement noise — it is essentially identical across tracks at a given altitude — so it carries no separable tracking-difficulty signal. *(b) The binding corner is non-reproducibly unstable.* At z=2 m and UGV speed 1.0 m/s — exactly where the coverage constraint binds hardest — identical configurations yield ArUco visibility from 35% to 100%, so no crisp feasibility boundary can be reported there. Establishing the isolated observability benefit would require decoupling the ground vehicle's navigation from the tracking camera (onboard odometry or a second sensor); we leave it as future work. [FUTURE WORK]

VIII. CONCLUSIONS

We studied aerial–ground cooperative tracking where a single downward UAV camera is the ground vehicle's sole exteroceptive sensor, serving both its navigation reference and its state estimate, in a genuinely-controlled, headless-reproducible ROS 2/Gazebo SITL system. We formalized the resulting coverage–resolution altitude tension, derived the perception-constrained altitude policy (fly as low as segmentation allows, climbing into turns), and showed a perception-driven event trigger is the computational enabler of the belief-space planner — $\approx 28\times$ fewer solves and $\approx 14\times$ less planner CPU at matched estimation quality, for a small loss of inter-trigger trajectory smoothness. We were explicit that the isolated observability benefit is not separable on this platform (the shared camera couples navigation and estimation) and that the binding low-altitude/high-speed corner is non-reproducibly unstable — both stated as findings, and both pointing to the same architectural cause. Future work: decouple the ground vehicle's navigation from the tracking camera (onboard odometry or a second sensor) to isolate the observability benefit; a full multicopter model for hardware-faithful altitude dynamics; and, on the theory side, a homothetic tube with recursive-feasibility guarantees.

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